

Original papers

Early detection of red palm weevil infestations using deep learning classification of acoustic signals

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ARTICLE INFO

Keywords:

Red Palm Weevil
Deep learning
Disease detection
Sound data
Smart agriculture

ABSTRACT

The Red Palm Weevil (RPW), also known as the palm weevil, is considered among the world's most damaging insect pests of palms. Current detection techniques include the detection of symptoms of RPW using visual or sound inspection and chemical detection of volatile signatures generated by infested palm trees. However, efficient detection of RPW diseases at an early stage is considered one of the most challenging issues for cultivating date palms. In this paper, an efficient approach to the early detection of RPW is proposed. The proposed approach is based on RPW sound activities being recorded and analyzed. The first step involves the conversion of sound data into images based on a selected set of features. The second step involves the combination of images from the same sound file but computed by different features into a single image. The third step involves the application of different Deep Learning (DL) techniques to classify resulting images into two classes: infested and not infested. Experimental results show good performances of the proposed approach for RPW detection using different DL techniques, namely MobileNetV2, ResNet50V2, ResNet152V2, VGG16, VGG19, DenseNet121, DenseNet201, Xception, and InceptionV3. The proposed approach outperformed existing techniques for public datasets.

1. Introduction

The red palm weevil (RPW), *Rhynchophorus ferrugineus*, is a significant pest of palm trees, causing considerable damage to the tree and resulting in substantial financial losses for farmers. RPW is considered among the most destructive pests that threaten palms worldwide. Saudi Arabia provides about \$2 million contribution to support the global efforts to eradicate RPW (Lütkebohle, 2018). However, several limitations still need to be addressed, especially on the farm side. In most cases, farm owners did not have any automated process providing clear indicators about infested palm trees, which causes from 25 to 40% loss of harvest due to unregulated processes (El-Juhany et al., 2010).

The management of the RPW has presented a significant challenge for entomologists since the beginning of modern agriculture, especially in the early detection of infested palms. Visual inspection of palms is the most adopted technique to locate infested palms. To detect palm diseases, farm owners would need to pay a significant amount of money to hire a palm tree infection expert for a whole year. Depending on the terrain and density, an expert can inspect around 200 palm trees daily. In many cases, there is up to 40% loss of crop yield due to the

inexperience of the farmworkers or unregulated processes. In addition, the problem of RPW detection becomes more difficult when it comes to early detection (Khan et al., 2022; Hu et al., 2022a). Indeed, RPW is a hidden pest for most of its life stages, except for the adult stage. Hence, it is imperative to ensure proper measures for early RPW detection. RPW causes various morphological changes to the palm tree, such as the formation of holes in the trunk and leaves, wilting, and discoloration. Infested palm trees often have a foul smell, and their fronds turn yellow and droop. In the later stages of infestation, the crown of the palm tree collapses, leading to the death of the entire tree. However, these morphological changes are challenging to detect in the early stages of infestation. Therefore, using an alternative technique, such as sound data, becomes crucial in detecting the presence of the RPW. Using sound data provides a non-invasive and efficient method for early detection of RPW infestation in palm trees, making it a valuable tool for farm owners and professionals in the agricultural industry.

For these reasons, there is a pressing need to use advanced information technologies in date farming. Deep learning (DL) has recently shown great promise in various applications, from natural language

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<https://doi.org/10.1016/j.compag.2023.108154>

Received 3 July 2023; Received in revised form 2 August 2023; Accepted 13 August 2023

Available online 28 August 2023

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processing and computer vision to healthcare and finance (Rehman et al., 2022; Ben Atitallah et al., 2022a; Boulila et al., 2022). DL models can learn complex patterns and relationships in data and make predictions or decisions based on that learned knowledge (Al-Sarem et al., 2021; Ben Atitallah et al., 2022b). These models can learn from large amounts of data and can improve their performance (Ghandorh et al., 2022). Due to these capabilities, DL is being used to solve many problems, including detecting the red palm weevil from sound data.

In this study, a novel approach for early disease detection is proposed. The proposed approach is a DL-based method for detecting infestation in palm trees using sound data. The process starts by recording sound data from palm trees and converting it into images by computing a set of selected features. The images are combined and classified using DL techniques to identify an infestation.

The main contributions of the proposed approach are summarized as follows:

- We proposed to combine three types of audio feature extraction techniques: Constant Q Cepstral Coefficients (CQCC) (Su et al., 2023), Mel-frequency Cepstral Coefficients (MFCC) (Abaskhah et al., 2023), and Bark Frequency Cepstral Coefficients (BFCC) (Gambhir et al., 2023). These features are combined to improve the classification accuracy of RPW infestation.
- We proposed an efficient method to detect early signs of RPW infestation, providing early intervention from farm owners and professionals.
- Several experiments are conducted using three public datasets: TreeVibe (Rigakis et al., 2021), Environmental Sound Classification (ESC) (Inik, 2023). Results show that the proposed approach provides excellent performances compared to existing related works.

The rest of the paper is organized as follows. Section 2 reviews relevant literature on the detection of palm tree infestation, highlighting key findings and gaps in the existing research. Section 3 details the study area, dataset description, and data collection. Section 4 describes the research design and analysis procedures used in the study. Results are presented in Section 5. In this section, the findings of the proposed study, including a comparison with existing techniques for public datasets. Section 6 interprets the results of the study in the context of the existing literature and provides insights into the implications of the findings. Finally, the main findings of the proposed study, their implications, and recommendations for future research are summarized in Section 7.

2. Related works

Recently, the application of new and modern agricultural techniques has proved its efficiency in many agriculture-related fields (Roopaei et al., 2017; Bu and Wang, 2019; Yang et al., 2021; Sinha and Dhanalakshmi, 2022). In the context of our study, we are interested in reviewing the use of modern technologies in the case of early disease detection, which is among the top priorities of farm stakeholders since it will ensure healthy farms and high-quality crops. In the literature, several works aim to use modern technologies to ensure the early detection of plant infestation (Hu et al., 2022b; Haridasan et al., 2023; Mallick et al., 2023; Yu et al., 2023; Zhu et al., 2023). However, this research area still needs more investigation, especially for palm trees (Ferreira et al., 2021; Goldshtein et al., 2022; Putra et al., 2022). In this paper, we have reviewed research works closely aligned with our research endeavors, which is the early detection of RPW infestation using acoustic signals. Koubaa et al. in Koubaa et al. (2020) proposed an IoT-based framework for red palm weevil early detection. The proposed framework allows the detection of red palm weevils by using smart agriculture sensors and collecting data using accelerometer sensors. Signal processing and probability methods are developed to identify the activity of the red palm weevil. Pinhas et al. (2008) developed a mathematical method

to automatically detect the acoustic signals of RPW larvae in palm offshoots using off-the-shelf recording devices. They utilized techniques similar to those used in speech recognition, such as vector quantization and Gaussian mixture modeling, to achieve detection ratios of 98.9%. This method could be used to monitor the trade and transportation of offshoots, helping to prevent the spread of this pest. In this paper (Parvathy et al., 2021), a Convolutional Autoencoder based Deep Learning model is proposed for identifying Red Palm Weevil (RPW) signals from background noise. The model uses Mel spectrogram images of RPW acoustic activities as the training data, achieving a classification accuracy of 95.85% on the test dataset composed of normal RPW acoustic emissions as well as anomalous acoustic samples. The proposed method is found to be highly efficient for the identification of RPW signals. Mohamed et al. (2021) proposed a method based on recording RPW sounds during different activity states, extracting important features in the sounds using the MFCC technique, and using artificial neural networks to automatically detect if RPW larvae exists. The recorded sounds were transformed into a spectrogram and analyzed with a CNN model, achieving 99.2% accuracy using AlexNet (Alom et al., 2018). This AI-based approach could help to detect RPW larvae early, potentially preventing the spread of this pest. In Wang et al. (2021), the authors introduced the combination of machine learning and fiber optic distributed acoustic sensing (DAS) techniques as a solution for the early detection of RPW in vast farms. They trained a fully-connected artificial neural network and a convolutional neural network using experimental time- and frequency-domain data provided by the fiber optic DAS system. The models can efficiently recognize healthy and infested trees with high classification accuracy values (99.9% by ANN with temporal data and 99.7% by CNN with spectral data, in reasonable noise conditions). This work paves the way for deploying the high-efficiency and cost-effective fiber optic DAS to monitor RPW in open-air and large-scale farms containing thousands of trees. In Ashry et al. (2022), the authors integrated optical fiber DAS and machine learning (ML) for the early detection of true weevil larvae less than three weeks old. They recorded temporal and spectral data using the DAS system and processed it with a 100–800 Hz filter to train CNN models that distinguish between “infested” and “healthy” signals with a classification accuracy of 97%. They also introduced a strict ML-based classification approach to improving the false alarm performance metric of the system. This approach could help to detect RPW early, preventing the spread of this harmful pest. In Esmail Karar et al. (2022), the authors proposed an IoT-based system for early detection of RPW larvae in date palms using a modified mixed depthwise convolutional network (MixConvNet) as a deep learning classifier. The proposed system was tested on the public TreeVibes dataset and achieved an accuracy score of 97.38%. The proposed system is considered a future milestone for practical implementation in the field.

Table 1 summarizes the discussed related works. As shown in the related works, most authors rely on spectral data in their research. In this paper, we present an innovative way to detect RPW based on the combination of features extracted from the collected sound data.

3. Dataset collection and description

Saudi Arabia is the second-largest palm date producer. Al-Ahssa region in Saudi Arabia was taken as a case study to test the performance of the proposed approach. The dataset used in this study originally is a set of recordings. The recordings were taken from date palm trees using remote vibroacoustic surveillance of trees. The device¹ used in this study is placed on the palm tree's trunk and records the vibrations created by the weevils as they chew and move inside the tree.

The reading points were marked on each date palm trunk (one meter above the ground level). Then drill holes (30 degrees) were made using

¹ <http://www.insectronics.net/>.

Table 1

Comparison of recent research works related to RPW detection using audio signals.

Ref.	Year	Algorithm	Data type	Steps	Dataset	Accuracy
Pinhas et al. (2008)	2008	GMM VQ	Spectral	- Record the RPW using off the shelf device - Manual data labeling - Calculate noise threshold - Counting the mismatched labels	–	0.988 0.989
Parvathy et al. (2021)	2021	CAE	Mel Spectrograms	- Capture audio signals - Extract features using Mel Spectrograms algorithm - Convert the features to images - Train CAE on the images	–	0.958
Mohamed et al. (2021)	2021	AlexNet	MFCC	- Extract features using MFCC algorithm - Convert the features to images - Train AlexNet on the images	–	0.992
Wang et al. (2021)	2021	ANN CNN	Temporal Spectral	- Install fiber optic DAS system on palm trees - Capture audio signals - Train ANN and CNN with DAS data - Deploy DAS system in large farms	–	0.999 0.997
Ashry et al. (2022)	2022	CNN	Temporal Spectral	- Record the sound of RPW using DAS - Transform it into temporal and spectral data - Train CNN on the data separately - Merge CNN models	–	0.970 0.971
Esmail Karar et al. (2022)	2022	CNN	Spectral	- Extract spectral features - Convert the features to images - Train MixConvNet on the images - Compare performance with other models	TreeVibes	0.973

Table 2

Data description.

Class	Number of samples
Infested	531
Not infested	575

a drill machine equipped with a brad point drill-bit (diameter, 8 mm) for a 35 cm depth inside the palm tree. A single hole was made on the site of the trunk, and an acoustic device covered the whole tree for RPW sound detection. For the reading, the acoustic device has a probe (waveguide) (35 cm in length) that is inserted inside the tree embedded with a recording machine. The waveguide was inserted into the hole of the date palm trunk that was previously drilled. The machine recording is connected to headphones that enable the user to assess the sound signal produced by RPW activities. This device provides the data which enables the movement from the device and the server/cloud system. The RPW device can detect the sound signal inside the date palm tree within a sphere of 1.5–2 meters radius (Rigakis et al., 2021). In this experiment, the parameter was set up to five minutes of repeat period and 20 s of recording times. Twenty seconds of sound data are contained in each tree recording daily. This record protocol was done in all date palm trees. Time stamps of each recording were also provided.

In order to better understand the dataset, Table 2 shows the number of recordings per class.

After collecting the data, several features are computed based on the sound dataset. Then, these features are combined to better describe the content of the data. The process of feature extraction and collaboration is described in the following section.

4. Proposed approach

The goal of this study is to detect RPW infestation from sound data acquired from palm trees. Fig. 1 describes the architecture of the proposed approach. The process starts by converting sound data into images by computing a set of selected features. The second step will combine images of the same sound file (computed by different features) into a single image. The third step involves applying different DL techniques to classify resulting images into two classes: infested and not infested.

4.1. Feature extraction from sound weevil dataset

We used CQCC, MFCC, and BFCC feature extractions to convert one sound file into three images, one for each feature.

To train DL models, extracting useful features from recorded sound data is required. In literature, several features have been investigated (Chen et al., 2021). A waveform can be converted to a spectrogram, which is a visual representation of all frequencies over time. Many features can be computed on sound data. In this study, we will consider MFCC, CQCC, and BFCC.

4.1.1. Mel-frequency cepstral coefficients (MFCC)

The cepstrum of a signal contains information about the rate of change in its spectral bands. A cepstrum is just a spectrum of the log of the temporal signal's spectrum. MFCCs are coefficients that make up the mel-frequency cepstrum (Zheng et al., 2001). The cepstrum communicates values that constitute a sound's formants (a distinguishing component of its quality) and timbre. As a result, MFCCs are very useful in DL models. Fig. 2 depicts the block diagram of MFCC (Ibrahim et al., 2008).

4.1.2. Constant Q cepstral coefficients (CQCC)

Combining the constant Q transform (CQT) and conventional cepstral processing results in the CQCC. CQT's design is appropriate for musical representation and transforms a data series to the frequency domain Brown (1991). It is a time–frequency analysis that perceptually motivated more than The short-term Fourier transform (STFT) (Allen, 1977). The ratio of the center frequency f_k to the bandwidth δf is known as the Q factor, which measures the selectivity of each filter.

$$Q = \frac{f_k}{\delta f} \quad (1)$$

The window function of the STFT is coupled to the constant bandwidth of each filter. Since all filters have the same absolute bandwidth f , the Q factor increases as one moves from low to high frequencies. Contrarily, the Q factor of the human sensory system is known to remain about constant between 500 Hz and 20 kHz (Moore, 2003). So, from a perceptual perspective, the STFT might not always be the optimum choice for the time–frequency analysis of speech signals.

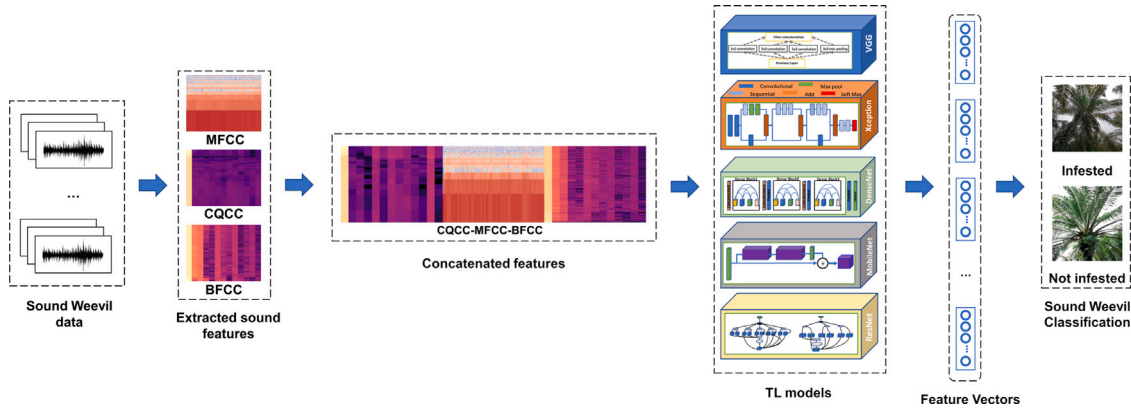


Fig. 1. Architecture of the proposed approach.

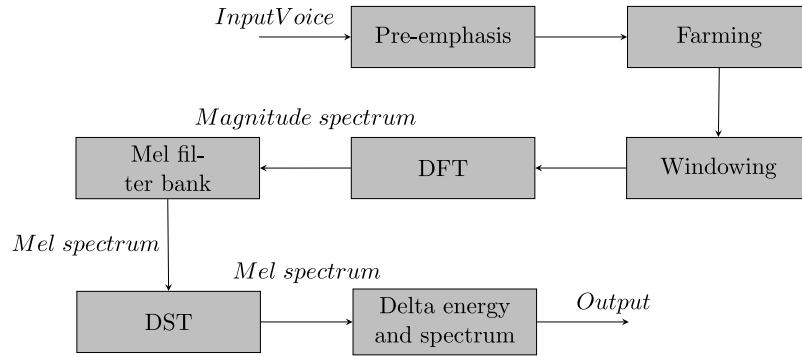


Fig. 2. MFCC block diagram.

The transform can be represented as a succession of filters f_k , each one k th having a spectral width δf_k that is a multiple of the width of the filter before it:

$$\delta f_k = 2^{1/n} \cdot \delta f_{k-1} = (2^{1/n})^k = \delta f_{min} \quad (2)$$

Where n is the number of filters per octave, f_{min} is the center frequency of the lowest filter, and δf_k is the bandwidth of the k th filter.

For CQT computation, the discrete time domain signal $x(n)$ has the following CQT $X^{CQ}(k, n)$ definition:

$$X^{CQ}(k, n) = \sum_{j=n-\lfloor N_k/2 \rfloor}^{n+\lfloor N_k/2 \rfloor} x(j) a_k^*(j - n + \frac{N_k}{2}) \quad (3)$$

where $a_k^*(n)$ is the complex number of the basis functions $a_k(n)$, K is the frequency bin index, $k = 1, 2, \dots, K$ and N_k are variable window lengths. The notation $\lfloor \cdot \rfloor$ denotes the floor function. The $a_k(n)$ functions are diverse time-frequency elements, defined as:

$$a_k(n) = \frac{1}{C} \left(\frac{n}{N_k} \right) \exp[i(2\pi n \frac{f_k}{f_s} + \phi_k)] \quad (4)$$

where k is a phase offset, f_k is the bin k 's center frequency, and f_s is the sampling rate. The scaling factor C can be found by using:

$$C = \sum_{l=-\lfloor N_k/2 \rfloor}^{\lfloor N_k/2 \rfloor} w \left(\frac{l + N_k/2}{N_k} \right) \quad (5)$$

where $w(x)$ represents a window function. The Eq. (6) then allows for the more-or-less traditional extraction of CQCC.

$$CQCC(p) = \sum_{l=1}^L \log |X^{CQ}(l)|^2 \cos \left[\frac{p(l - \frac{1}{2})\pi}{L} \right] \quad (6)$$

where l are the recently resampled frequency bins and $p = 0, 1, \dots, L - 1$ (see Fig. 3).

4.1.3. Bark frequency cepstral coefficients (BFCC)

BFCC is a feature extraction technique used in speech and audio processing. It is based on the Bark scale, which is a non-linear frequency scale that approximates the human auditory system's response to different frequency bands. The Bark scale is defined as:

$$z = 26.81 * (f/1000) - 0.53 * (f/1000)^2 + (4.5 * 10^{-6}) * (f/1000)^3 \quad (7)$$

Where f is the frequency in Hz, the BFCCs are calculated by first converting the audio signal into the Bark scale, applying the above formula on the frequency axis of the spectrogram of the audio signal, then taking the cepstral coefficients of the resulting Bark-scale spectrogram. The cepstral coefficients are obtained by applying a Discrete Cosine Transform (DCT) on the log-magnitude of the Bark-scale spectrogram. This process is known as Cepstral Analysis. The result is a set of coefficients that represents the audio signal in the Bark frequency scale. These coefficients are often used as input to machine learning models for tasks such as speech recognition and music classification.

4.2. Feature combination

DL techniques have the ability to learn representations of data using multiple abstraction levels. The feature combination aims to take advantage of multiple features to construct a robust discriminator of red weevil sound. Each computed feature determines a specific aspect of the audio signal's temporal or spectral content. Combining these features will help the DL techniques discriminate between infested and not infested palms. Using the early mentioned features, DL models can be trained separately using only one feature at a time or combined features.

Algorithm 1 describes the detailed process of combining the extracted features using our selected feature extractions CQCC, MFCC, and BFCC, where lines 2 to 4 specify the static variables such as data

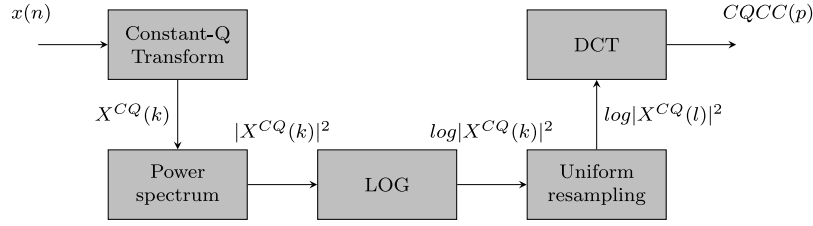


Fig. 3. CQCC block diagram.

paths and feature extractions. Lines 6 and 7 are for audio file loading based on the current file name on the loop at line 5. After loading our audio file, in lines 10 to 12, we extract the audio features and convert it to an image without any axis. In line 13, we append the generated image in a temporary list that clears itself each time after loading an audio file. After extracting the audio feature and appending the images to our temporary list, lines 15 to 17 combine the generated images and save them as one image with the same name as the audio file. This process will be repeated on all the audio files in the target folder.

Algorithm 1 : Combine multiple feature extraction methods

```

1: Begin
2:  $audioFolderPath \leftarrow /home/data/audio$ 
3:  $imagesFolderPath \leftarrow /home/data/images$ 
4:  $extractionMethods \leftarrow [cqcc, mfcc, bfcc]$   $\triangleright$  specify the extraction methods
5: for each  $fileName \in audioFolderPath$  do  $\triangleright$  loop over all files in the audio folder
6:    $audioFilePath \leftarrow joinPath(audioFolderPath, fileName)$ 
7:    $audio = loadAudio(audioFilePath)$ 
8:    $tempImagesList \leftarrow \emptyset$ 
9:   for each  $method \in extractionMethods$  do
10:     $features \leftarrow method.extract(audio)$   $\triangleright$  use the current algorithm for the extraction
11:     $figure = createFigure()$   $\triangleright$  create empty figure
12:     $figure.display(features)$   $\triangleright$  display the extracted features on the figure
13:     $tempImagesList.append(figure)$   $\triangleright$  save the features image temporarily
14:   end for
15:    $combinedImage \leftarrow concat(tempImagesList)$   $\triangleright$  concatenate features images
16:    $imageFilePath \leftarrow joinPath(imagesFolderPath, fileName)$ 
17:    $save(imageFilePath, combinedImage)$ 
18: end for
19: End

```

4.3. DL classification of sound weevil

We applied several DL techniques, namely MobileNetV2, ResNet-50V2, ResNet152V2, VGG16, VGG19, DenseNet121, DenseNet201, Xception, and InceptionV3 on the generated Red Palm Weevil dataset. We used TensorFlow and Keras frameworks to train the listed TL models on 200 epochs, 0.0001 learning rate, and RMSprop optimizer. The dataset has been split into three sets, training set (0.8), validation set (0.1), and test set (0.1), to ensure preventing any results coming from overfitting.

These techniques are used to extract features from an audio dataset, concatenate the CQCC, MFCC, and BFCC images of each audio file together, and train models to classify the audio samples. CQCC, MFCC, and BFCC are feature extraction techniques commonly used in audio processing to extract different audio characteristics. By combining these features in the form of images and training CNNs on them, the models can learn and identify complex patterns in the audio.

These specific CNNs are chosen based on their performance and efficiency in the image and audio classification tasks. MobileNetV2 and Xception are lightweight models suitable for mobile and embedded vision applications, while ResNet50V2 and ResNet152V2 are variants of the ResNet architecture that improve accuracy by using residual connections and bottleneck layers. VGG16, VGG19, DenseNet121, DenseNet201, and InceptionV3 are also widely used CNNs that have been proven to be effective in various image and audio classification tasks. RMSprop optimizer was used for all training experiments as it is particularly well-suited for training deep neural networks with a large number of parameters.

Using a variety of CNNs in this study demonstrates the versatility and applicability of the proposed approach in various audio classification tasks and with different types of CNN architectures. The combination of CQCC, MFCC, and BFCC features with various CNNs allows the extraction of different audio characteristics and enhances the models' performance.

5. Results

5.1. Feature extraction from the sound weevil dataset

Feature extraction is a process in which characteristics or features of an audio signal are extracted and represented in a more manageable and informative way. It is crucial in many audio processing tasks, such as speech recognition, music classification, and sound event detection. The extracted features serve as input to machine learning models, which use them to make predictions or decisions. There are many different feature extraction techniques that can be used, each with its own strengths and weaknesses. The choice of feature extraction method depends on the specific task and the characteristics of the audio signal. Fig. 4 illustrates the use of three different feature extraction techniques, namely CQCC, MFCC, and BFCC, on a dataset of audio files labeled as "infested" or "not infested". The figure contains four samples from each class, and the output of the three feature extraction techniques is displayed for each sample.

5.2. Red palm weevil detection

In this section, we present the validation accuracy results of transfer learning techniques with nine backbone models, namely MobileNetV2 (Sandler et al., 2018), ResNet50V2 (He et al., 2016a), ResNet152V2 (He et al., 2016b), VGG16 (Simonyan and Zisserman, 2014a), VGG19 (Simonyan and Zisserman, 2014b), DenseNet121 (Huang et al., 2017a), DenseNet201 (Huang et al., 2017b), Xception (Chollet, 2017), and InceptionV3 (Christian Szegedy et al., 2015), for each feature extracted from the sound Weevil dataset as well as the combination of these features. The primary purpose of applying different DL techniques is to check the performance of the combination of selected sound features in detecting the RPW infestation.

Table 3 provides a comparison of the performance of nine different machine learning models: MobileNetV2, ResNet50V2, ResNet152V2, VGG16, VGG19, DenseNet121, DenseNet201, Xception, and InceptionV3. These models are compared based on their top-1 accuracy

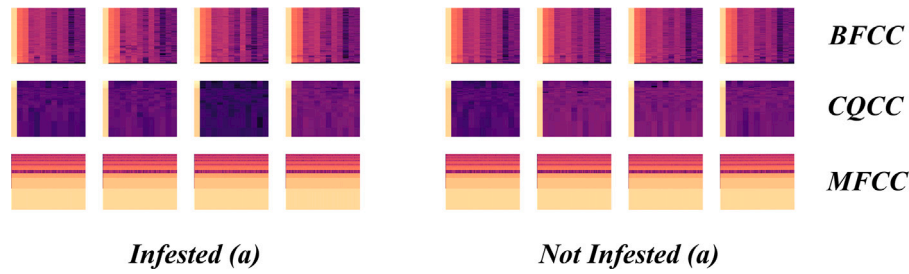


Fig. 4. The resulting images for each class.

Table 3

Comparative analysis of MobileNetV2, ResNet50V2, ResNet152V2, VGG16, VGG19, DenseNet121, DenseNet201, Xception, and InceptionV3 models on top-1 accuracy, top-5 accuracy, number of parameters, and GFLOPS using the ImageNet dataset.

Model	Accuracy@1	Accuracy@5	Parameters (in millions)	GFLOPS
MobileNetV2 (Sandler et al., 2018)	72.154	90.822	3.5M	0.3
ResNet50V2 (He et al., 2016a)	80.858	95.434	25.6M	4.09
ResNet152V2 (He et al., 2016b)	82.284	96.002	60.2M	11.51
VGG16 (Simonyan and Zisserman, 2014a)	71.592	90.382	138.4M	15.47
VGG19 (Simonyan and Zisserman, 2014b)	72.376	90.876	143.7M	19.63
DenseNet121 (Huang et al., 2017a)	74.434	91.972	8.0M	2.83
DenseNet201 (Huang et al., 2017b)	76.896	93.37	20.0M	4.29
Xception (Chollet, 2017)	79.0	94.5	22.9M	–
InceptionV3 (Christian Szegedy et al., 2015)	77.294	93.45	27.2M	5.71

(Accuracy@1), top-5 accuracy (Accuracy@5), the number of parameters (in millions), and their computational complexity in GFLOPS (Giga Floating Point Operations Per Second). The accuracies are evaluated on the ImageNet dataset and represent the model's ability to correctly identify the true class out of its top 1 and top 5 predictions, respectively. The number of parameters in the model represents the model's size and capacity, and GFLOPS is a measure of the computational complexity of the model, with higher values indicating more computation is required. The exact values for each model will need to be filled in according to the specific experiment results.

Table 4 shows the accuracy and loss values for the classification experiments using the mentioned models. We used single-feature extraction algorithms to compare the classification performances of the considered DL models. The features depicted in the table are LFCC, CQCC, MFCC, Chroma, GFCC, BFCC, Mel Spectro, and Spectral Centroid. The highest accuracy was achieved by the models trained on the CQCC feature set, with a maximum accuracy of 1.000 using the DenseNet121 model. Achieving 100% accuracy using only one feature, CQCC, is not sufficient because we aim to achieve the best accuracy regardless of the CNN architecture used. Therefore, we used multiple feature combinations to ensure high accuracy without relying on any specific CNN architecture. On the other hand, the lowest accuracy was achieved by the models trained on the MFCC feature, with a minimum accuracy of 52% using the ResNet152V2 model.

In Table 5, several experiments have been conducted with multiple feature combinations, including CQCC-MFCC-BFCC, MFCC-Chroma-Mel Spectro, MFCC-Chroma-LFCC, GFCC-BFCC-CQCC, GFCC-CQCC-LFCC, and Spectro-MFCC-BFCC. The accuracy values in the table ranged from 70% to 100%, indicating that the models performed very well on the test data. The loss values were all relatively low, ranging from 0.000 to 9.836. A lower loss value indicates that the model is better at predicting the target variable. It is interesting to note that all models trained on the CQCC-MFCC-BFCC feature set achieved perfect accuracy, while some of the models trained on other feature sets achieved lower accuracy values. The ResNet152V2 and DenseNet201 models performed very well across all feature sets. Table 5 highlights that the choice of a specific audio feature set has a significant impact on the accuracy of the models. It also suggests that the choice of model architecture is important for achieving good accuracy in the audio classification task.

Fig. 5 shows the validation accuracy and loss progress of all the used DL models during the training; each line consists of 200 points, one for each epoch.

Fig. 6 presents a clearer version of Fig. 5, showing how fast the models are learning during the first 10 epochs.

Based on Figs. 5 and 6, it appears that all of the models were able to reach a validation accuracy of 100% within the first 10 epochs when using the merged features. The full distribution plot provides a more comprehensive view of the models' performance throughout the training. In contrast, the plot of the first 10 epochs provides a more focused view of the models' initial performance. This confirms that all models were able to effectively learn and make accurate predictions on the validation set within a relatively short period of time. One possible explanation for this is that the models were pre-trained on a large dataset, which allowed them to effectively learn the features of the validation set in a relatively short time. Additionally, the high accuracy achieved within the first 10 epochs also suggests that the models were able to effectively generalize to unseen data, which is an important characteristic of a good model.

The confusion matrix results presented in Fig. 7 are impressive, as they show that all of the provided models were able to achieve 100% accuracy on the test set. This suggests that the models have been effectively trained and are able to make accurate predictions on unseen data, which is a key indicator of a high-performing model.

6. Evaluation

In this section, several experiments have been conducted to evaluate the performances of the proposed approach using three public datasets. The first dataset, TreeVibes,² contains recordings of internal vibrations of trees that are wirelessly transmitted to a cloud server using the device detailed in Section 3. The dataset aims to detect the presence of wood-boring insects that feed or move inside trees, which can cause high tree mortality. The dataset contains almost 1000 “clean” label recordings and over 50 000 “infested” label recordings, making it a valuable resource for detecting tree infestations and preventing damage caused by insect feeding.

² <https://www.kaggle.com/datasets/potamitis/treevibes>.

Table 4
Single feature experimental results.

Features	Model	Accuracy	Loss	Features	Model	Accuracy	Loss
LFCC	MobileNetV2	0.843	9.715	GFCC	MobileNetV2	0.848	7.609
	ResNet50V2	0.776	1.873		ResNet50V2	0.786	5.045
	ResNet152V2	0.734	2.385		ResNet152V2	0.864	1.679
	VGG16	0.765	1.288		VGG16	0.843	0.555
	VGG19	0.734	0.498		VGG19	0.828	0.496
	DenseNet121	0.739	3.185		DenseNet121	0.838	2.664
	DenseNet201	0.651	2.771		DenseNet201	0.817	2.318
	Xception	0.760	8.531		Xception	0.807	7.971
CQCC	InceptionV3	0.765	3.113	BFCC	InceptionV3	0.833	4.667
	MobileNetV2	0.994	0.008		MobileNetV2	0.791	7.068
	ResNet50V2	0.979	0.109		ResNet50V2	0.796	4.732
	ResNet152V2	0.979	0.161		ResNet152V2	0.723	7.052
	VGG16	0.994	0.017		VGG16	0.786	0.470
	VGG19	0.989	0.019		VGG19	0.770	0.499
	DenseNet121	1.000	0.000		DenseNet121	0.812	2.268
	DenseNet201	0.984	0.035		DenseNet201	0.718	3.300
MFCC	Xception	1.000	0.000	Mel Spectro	Xception	0.770	9.317
	InceptionV3	0.994	0.012		InceptionV3	0.791	2.716
	MobileNetV2	0.817	7.428		MobileNetV2	0.838	9.244
	ResNet50V2	0.848	1.549		ResNet50V2	0.833	1.923
	ResNet152V2	0.557	14.10		ResNet152V2	0.843	3.693
	VGG16	0.791	0.498		VGG16	0.765	0.396
	VGG19	0.520	0.692		VGG19	0.875	0.315
	DenseNet121	0.869	1.374		DenseNet121	0.791	3.158
Chroma	DenseNet201	0.807	4.450	Spectral Centroid	DenseNet201	0.859	2.950
	Xception	0.864	3.286		Xception	0.828	4.113
	InceptionV3	0.880	1.264		InceptionV3	0.802	1.753
	MobileNetV2	0.781	7.462		MobileNetV2	0.854	8.867
	ResNet50V2	0.750	2.337		ResNet50V2	0.854	2.005
	ResNet152V2	0.781	3.758		ResNet152V2	0.822	1.076
	VGG16	0.776	0.777		VGG16	0.859	0.273
	VGG19	0.677	0.620		VGG19	0.817	0.426
Chroma	DenseNet121	0.734	3.294	Spectral Centroid	DenseNet121	0.869	1.361
	DenseNet201	0.791	5.286		DenseNet201	0.869	1.479
	Xception	0.817	4.872		Xception	0.875	2.733
	InceptionV3	0.697	3.928		InceptionV3	0.859	3.972

Table 5
Combined features experimental results.

Features	Model	Accuracy	Loss	Features	Model	Accuracy	Loss
CQCC	MobileNetV2	1.000	0.000	GFCC	MobileNetV2	0.890	9.393
	ResNet50V2	1.000	0.000		ResNet50V2	0.854	3.961
	ResNet152V2	1.000	0.000		ResNet152V2	0.885	9.836
	VGG16	1.000	0.000		VGG16	0.854	0.355
MFCC	VGG19	1.000	0.000	BFCC	VGG19	0.760	0.481
	DenseNet121	1.000	0.000		DenseNet121	0.828	1.048
BFCC	DenseNet201	1.000	0.000	CQCC	DenseNet201	0.859	1.594
	Xception	1.000	0.000		Xception	0.848	9.100
	InceptionV3	1.000	0.000		InceptionV3	0.781	4.304
MFCC	MobileNetV2	0.875	6.764	GFCC	MobileNetV2	0.979	0.487
	ResNet50V2	0.786	9.704		ResNet50V2	0.838	3.509
	ResNet152V2	0.875	2.720		ResNet152V2	0.979	0.141
	VGG16	0.833	0.388		VGG16	0.786	3.453
Chroma	VGG19	0.817	0.548	CQCC	VGG19	0.708	0.475
	DenseNet121	0.838	1.894		DenseNet121	0.963	0.294
Mel Spectro	DenseNet201	0.833	1.945	LFCC	DenseNet201	0.989	0.229
	Xception	0.890	2.618		Xception	0.864	5.917
	InceptionV3	0.796	5.233		InceptionV3	0.828	2.079
	MobileNetV2	0.796	0.764		MobileNetV2	0.864	6.603
MFCC	ResNet50V2	0.807	3.938	Mel Spectro	ResNet50V2	0.833	2.246
	ResNet152V2	0.807	1.653		ResNet152V2	0.869	1.940
	VGG16	0.708	0.541		VGG16	0.828	0.461
	VGG19	0.817	1.222		VGG19	0.765	0.554
Chroma	DenseNet121	0.812	1.969	MFCC	DenseNet121	0.848	2.250
	DenseNet201	0.807	4.927		DenseNet201	0.822	2.373
	Xception	0.796	4.091		Xception	0.838	5.584
	InceptionV3	0.833	3.257		InceptionV3	0.807	1.684

We applied the InceptionV3 model with various features, including CQCC, MFCC, BFCC, Spectral-Centroid (Esmail Karar et al., 2022) and the proposed combined features (CQCC-MFCC-BFCC). The results

showed a 100% test accuracy in all the cases, indicating that the dataset may be too simple for DL methods. Since this evaluation section aims to assess the effectiveness of combining features in improving the

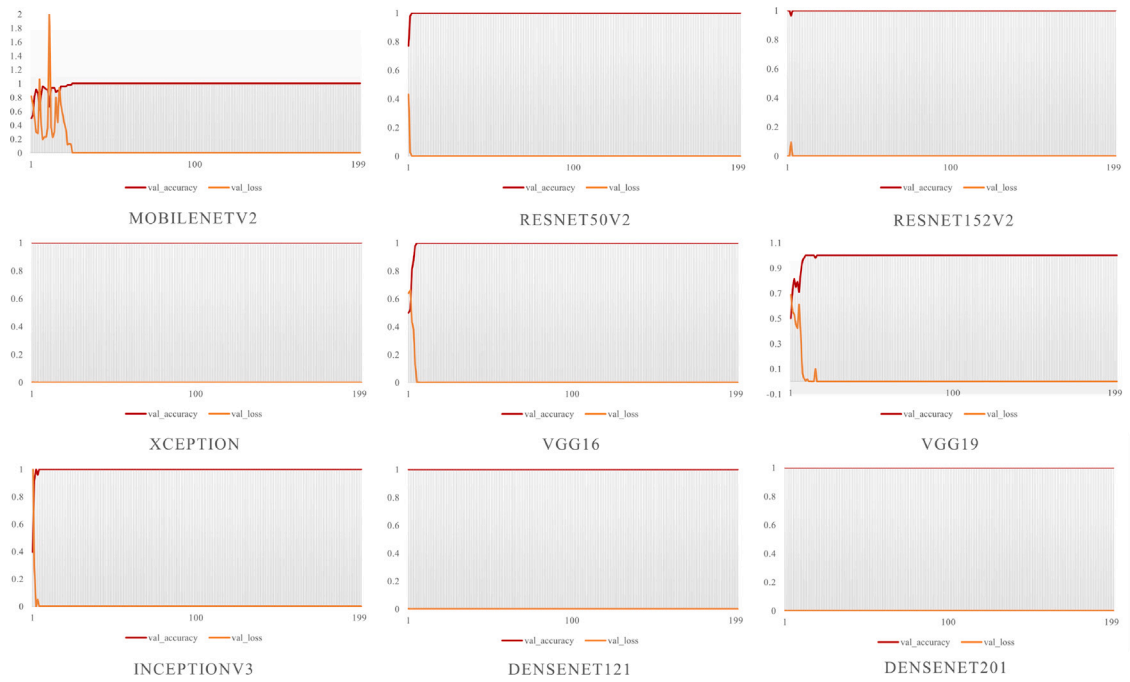


Fig. 5. Validation accuracy and loss progress of CQCC-MFCC-BFCC on RPW dataset.

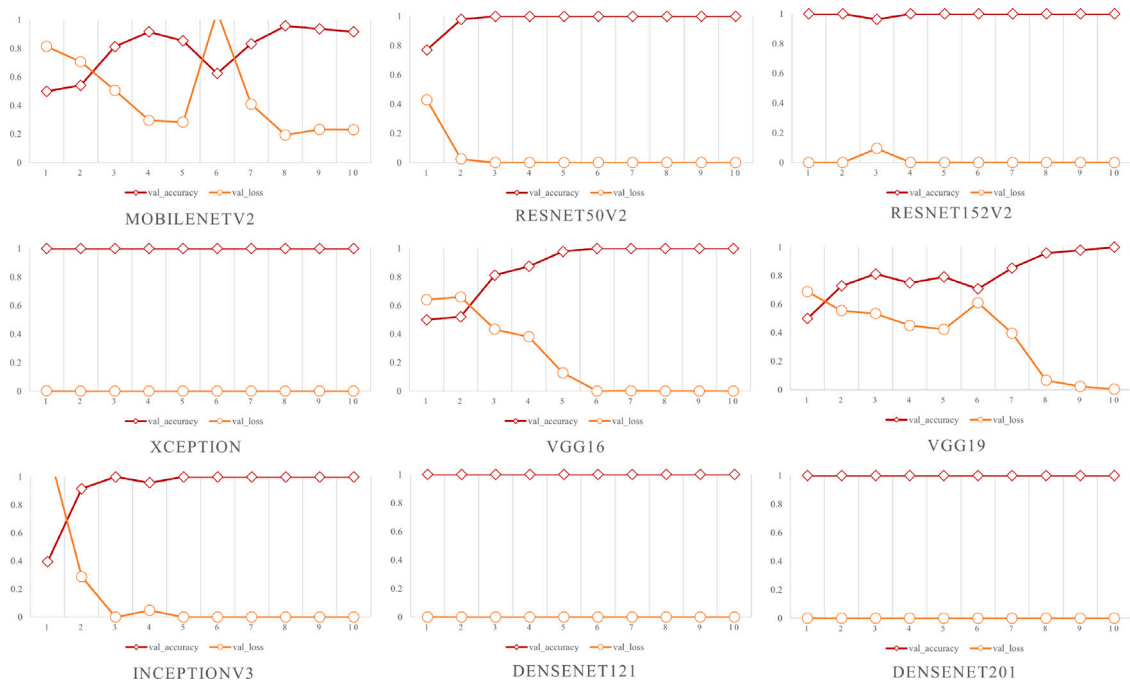


Fig. 6. Validation accuracy and loss progress of CQCC-MFCC-BFCC on RPW dataset for the first 10 epochs.

accuracy of detecting tree infestations, we will use machine learning techniques algorithms such as Support Vector Machine (SVM), Logistic Regression (LR), Decision Trees (DT), and Random Forest (RF). The results showed that using different features, the accuracy ranged from 0.980 to 1.000, with the best accuracy achieved using the proposed combined features with the LR method. Table 6 shows the detailed results of the experiments on the TreeVibes dataset. We note that the combination of the three features (CQCC-MFCC-BFCC) provides the

best accuracy of detecting tree infestations for all four machine learning techniques.

To better validate the performance of the proposed approach, we used two public datasets related to ESC, ESC-10³ and ESC-50.⁴ The two

³ <https://github.com/karolpiczak/ESC-10>.

⁴ <https://github.com/karolpiczak/ESC-50>.

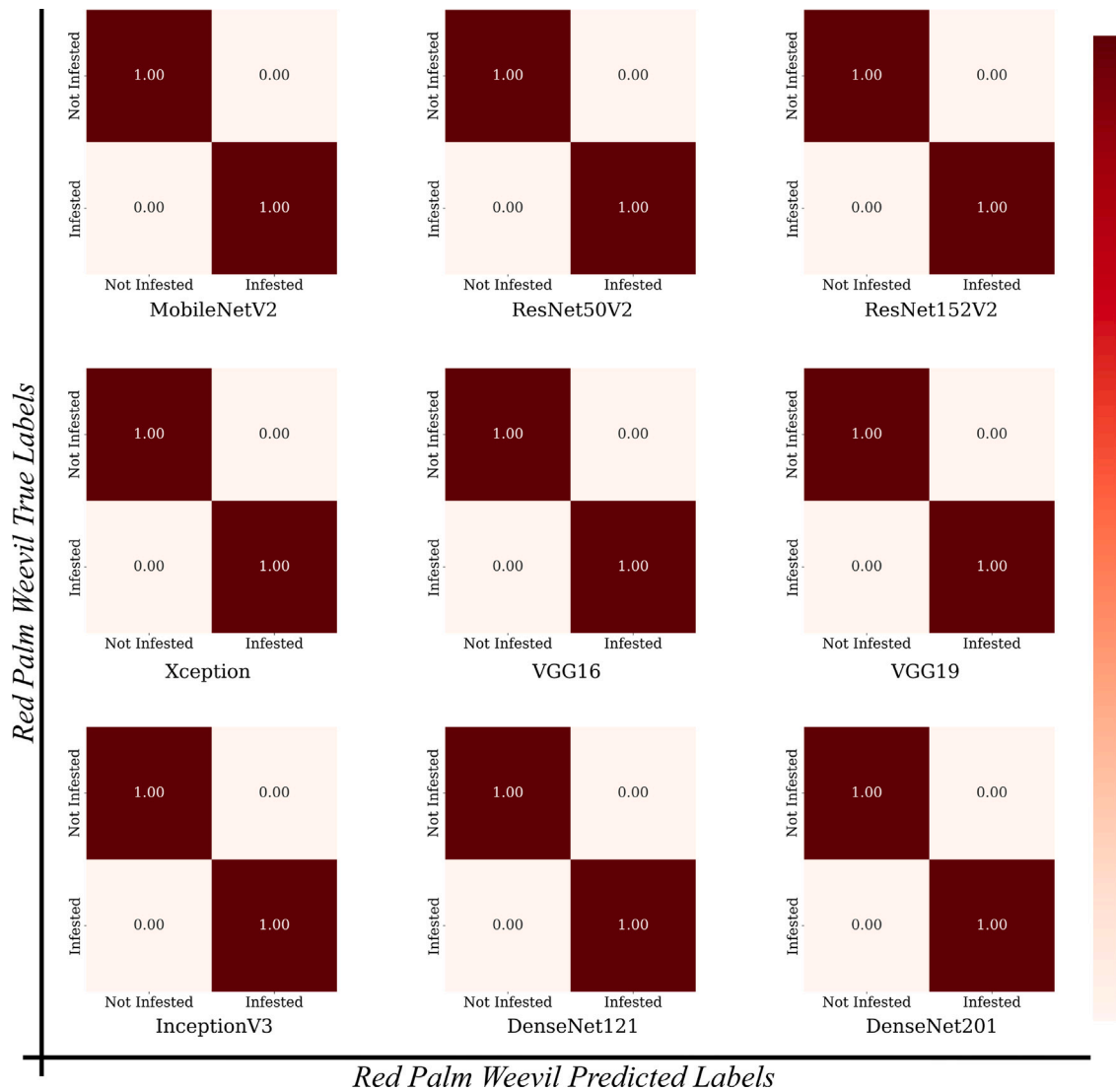


Fig. 7. Confusion matrices of CQCC-MFCC-BFCC on RPW dataset.

Table 6
Experimental results on Treevibes dataset.

Features	SVM	LR	DT	RF
CQCC	0.980	0.992	0.980	0.990
MFCC	0.982	0.998	0.990	0.997
BFCC	0.982	0.992	0.980	0.986
Esmail Karar et al. (2022)	0.995	0.995	0.995	0.995
Proposed approach	0.997	1.000	0.995	0.997

datasets consist of several brief environment recordings. Each recording consists of five seconds. We split the ESC-10 and ESC-50 datasets into train:val:test splits with ratios of 0.8:0.1:0.1.

For the ESC-10 dataset, we trained the InceptionV3 model on five different feature extraction techniques, including CQCC, MFCC, BFCC, LBP-VAR-ELBP-LBP1D-LPQ1D-MFCC-GFCC (Toffa and Mignotte, 2020), and our proposed combination of CQCC-MFCC-BFCC. Results showed that the proposed combination achieved the highest validation accuracy of 0.95 with a precision of 0.947, recall of 1.0, and an F1 score of 0.973. The second-highest accuracy of 0.925 was achieved by the LBP-VAR-ELBP-LBP1D-LPQ1D-MFCC-GFCC combination. The accuracies of the models trained on CQCC, MFCC, and BFCC were 0.875, 0.875, and 0.825, respectively. Table 7 shows the detailed results of the experiments on ESC-10 dataset.

Similarly, for the ESC-50 dataset, we trained the InceptionV3 model using the same five feature extraction techniques. The proposed combination of CQCC-MFCC-BFCC achieved the highest accuracy of 0.995 with a precision of 0.995, a recall of 1.0, and an F1 score of 0.997. The InceptionV3 model trained using the LBP-VAR-ELBP-LBP1D-LPQ1D-MFCC-GFCC (Toffa and Mignotte, 2020) combination achieved the second-highest accuracy of 0.985, with a precision of 0.995, recall of 0.99, and an F1 score of 0.992. The models trained using CQCC, MFCC, and BFCC achieved accuracies of 0.975, 0.96, and 0.98, respectively. Table 8 shows the detailed results of the experiments on the ESC-50 dataset. Based on these results, we can conclude that the proposed combination of CQCC-MFCC-BFCC achieved the highest accuracy for both ESC-10 and ESC-50 datasets.

The InceptionV3 model trained using the LBP-VAR-ELBP-LBP1D-LPQ1D-MFCC-GFCC combination (Toffa and Mignotte, 2020) also performed well for both datasets, achieving the second-highest accuracy. Moreover, the InceptionV3 model is a fast model, making it a good choice for these datasets. It is important to note that the choice of feature extraction technique could impact the final accuracy results, and the learning rate used during training could also affect the performance of the model.

Acoustic-based detection systems can complement image-based techniques for detecting plant diseases and infestations, especially

Table 7

Experimental results on the ESC10 dataset.

Features	Accuracy	Precision	Recall	F1 Score
CQCC	0.875	0.897	0.972	0.933
MFCC	0.875	0.919	0.944	0.932
BFCC	0.825	0.892	0.917	0.904
Toffa and Mignotte (2020)	0.925	0.971	0.944	0.958
Proposed approach	0.950	0.947	1.000	0.973

Table 8

Experimental results on ESC50 dataset.

Features	Accuracy	Precision	Recall	F1 Score
CQCC	0.975	0.995	0.980	0.987
MFCC	0.960	0.990	0.969	0.979
BFCC	0.980	0.990	0.990	0.990
Toffa and Mignotte (2020)	0.985	0.995	0.990	0.992
Proposed approach	0.995	0.995	1.000	0.997

when external symptoms are not apparent. In many situations, sound can be affected by background noise. Hence, DL techniques constitute an excellent solution since they are very effective in detecting RPW, even in background noise. This is because they can learn to distinguish between the characteristics of the RPW's acoustic signature and the noise. The acoustic signals of the RPW have a unique frequency range, waveform pattern, and duration that differ from those of other sounds that occur in the environment. DL models, such as convolutional neural networks (CNNs), can be trained to recognize these unique characteristics and distinguish them from background noise. However, it is essential to note that certain limitations need to be addressed to implement the proposed solution in a real-world scenario. First, we acknowledge that the TreeVibe device incurs a high cost for acquisition and maintenance, which may pose a challenge for widespread adoption. Additionally, the considered device is susceptible to noise if not correctly fitted or calibrated, which can impact the accuracy and reliability of the detection results. The proposed solution can be integrated into an architecture based on Optical Fiber Distributed Acoustic Sensing, as proposed in the paper (Ashry et al., 2022). Identifying the locations where the devices will be deployed is crucial, and this will involve mapping the farm and identifying the areas where RPW infestations are most prevalent.

7. Conclusions

Early detection of RPW infestation can play a crucial role in reducing the negative impact of this pest on palm trees and the industries that rely on them. Early detection can help to reduce the spread of the infestation, as prompt action can be taken to contain and control the population. This can prevent the red weevil from spreading to other trees, which can help to preserve the overall health of the palm population. Additionally, early detection can also help to reduce the cost of control measures. By taking action before significant damage occurs, less extensive and costly control methods may be required. This can also help to reduce the overall economic impact of red weevil infestations on the palm industry. This study suggests using a DL approach to detect RPW infestation in palm tree farms. The proposed approach utilizes three features, namely CQCC, MFCC, and BFCC, to convert sound data received from sensors embedded in palm trees into images. The classification of resulting images into two classes (infested and not infested), based on several DL techniques, shows good performances of the proposed approach on a dataset representing the Al-Ahssa region of the Kingdom of Saudi Arabia, as well as public datasets. The proposed approach can help to improve the accuracy and efficiency of detecting and managing RPW. This study can be extended considering several perspectives. First, the proposed system can be integrated with existing farm management tools to enable automated pest control and monitoring. In addition, an important

research area to explore will be the investigation of other advanced DL techniques, such as reinforcement learning or generative models, to improve the performance of the detection system and to predict the severity of the infestation and the potential damage to the tree. Finally, this paper can be extended by conducting more field trials in various settings. This could involve testing the system's effectiveness in different geographical locations, under varying climatic conditions, and with different species of palm trees, which could vary in their acoustic properties. Additionally, the impact of various background noises that might be present in recordings can be further explored. Such extensive field trials would provide a more robust understanding of the system's performance across diverse real-world conditions.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors would like to express their deepest gratitude to Prof. Abdulrahman S. Aldawood for providing the dataset. Also, the authors would like to thank Prince Sultan University for their support.

Funding

This work was supported by a research grant from Prince Sultan University, Saudi Arabia (seed-ccis-101)

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